Rotation Invariant and Edge Aware Mamba for Oriented Remote Sensing Object Detection

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***Abstract*—Oriented remote sensing object detection (oriented RSOD) extends general remote sensing object detection (RSOD) by localizing aerial objects with rotated bounding boxes, which is essential for dense and arbitrarily oriented targets in overhead imagery. High-resolution remote sensing scenes require broad contextual reasoning, yet CNN backbones are limited by local receptive fields and Transformer backbones incur quadratic cost in global token interactions. Mamba offers an efficient alternative because its state-space sequence modeling captures long-range dependencies with linear complexity. However, vanilla visual Mamba still lacks the geometric, boundary, and scale-aware priors required by oriented RSOD. To address this gap, we propose RIEMamba, a Mamba-based backbone that preserves efficient long-sequence modeling while embedding remote-sensing-specific priors. First, Rotation-Invariant Edge-Aware Pixel Difference Convolution (RIEPDC) integrates gradient-based operators into pixel-difference convolution with SO(2)-equivariant structural modeling to enhance rotation-consistent boundary representation. Second, Variable Multi-Scale Scanning (VMSS) fuses scanning information from two scale windows to balance local detail capture with contextual modeling. Experiments show that RIEMamba achieves state-of-the-art results on DOTA-v1.0 (80.11%/82.49% mean Average Precision (mAP) for single/multi scale), HRSC2016 (98.62% mAP under VOC 2012), and DIOR-R (68.55% mAP), with only slight computational overhead.**

***Index Terms*—Contextual information, deep neural networks, long-distance, mamba, remote sensing object detection (RSOD).**

I. Introduction

THE rapid advancement of deep learning has significantly driven progress in object detection for remote sensing imagery [1], [2], with most detection architectures directly borrowing design paradigms from general computer vision models [3], [4], [5], [6], [7]. Oriented remote sensing object detection (oriented RSOD) extends general remote sensing object detection (RSOD) by predicting rotated bounding boxes, which is necessary because aerial objects are often densely arranged, small, weakly bounded, and distributed at arbitrary orientations. As shown in Fig. 1(a), the DOTA dataset exhibits a highly imbalanced angular distribution, with several dominant peaks around 0°, 90°, 180°, and 270°, while many oblique angles contain far fewer instances. This long-tailed rotation pattern makes stable orientation-aware representation especially important.

When general visual architectures are transferred to oriented RSOD, three limitations become prominent. CNN backbones are efficient and strong at local texture extraction, but their local receptive fields make it difficult to capture broad contextual layouts in high-resolution aerial images. Transformer backbones provide global token interaction, but dense self-attention has quadratic complexity and becomes costly for large remote sensing images. Existing rotation-oriented strategies, such as data augmentation [8], [9], rotated anchors [10], [11], angle regression, or post-processing refinements [12], can improve oriented localization, but they often rely on external geometric handling rather than building rotation-consistent, edge-aware, and scale-adaptive features inside the backbone. Such handling mainly changes the training samples, anchor geometry, angle labels, decoded boxes, or NMS rules after feature extraction, so it cannot repair weak feature evidence produced earlier by an orientation-sensitive backbone. In dense aerial scenes, this means that two rotated instances with similar appearances may still produce inconsistent activations, weak object boundaries may still be mixed with background responses, and small targets may already be suppressed before angle correction or post-processing is applied. In addition, preset angle bins, anchor shapes, and refinement heuristics introduce quantization and hyperparameter sensitivity, which limits their adaptability to cluttered backgrounds, weak boundaries, and strong scale variation.

Fig. 1(b)–(d) further illustrates how these challenges appear in detection outputs and feature responses. The odd columns show detection results and the even columns show corresponding heatmaps; in each group, the top row is the baseline and the bottom row is RIEMamba. In the rotation case (b), the baseline misses or incompletely localizes several obliquely parked vehicles near the curved road, and its heatmap is dominated by the road texture rather than the vehicle bodies. RIEMamba recovers more rotated vehicle boxes and produces stronger activations aligned with the vehicles and lane-side structures, showing improved rotation-consistent response. In the edge

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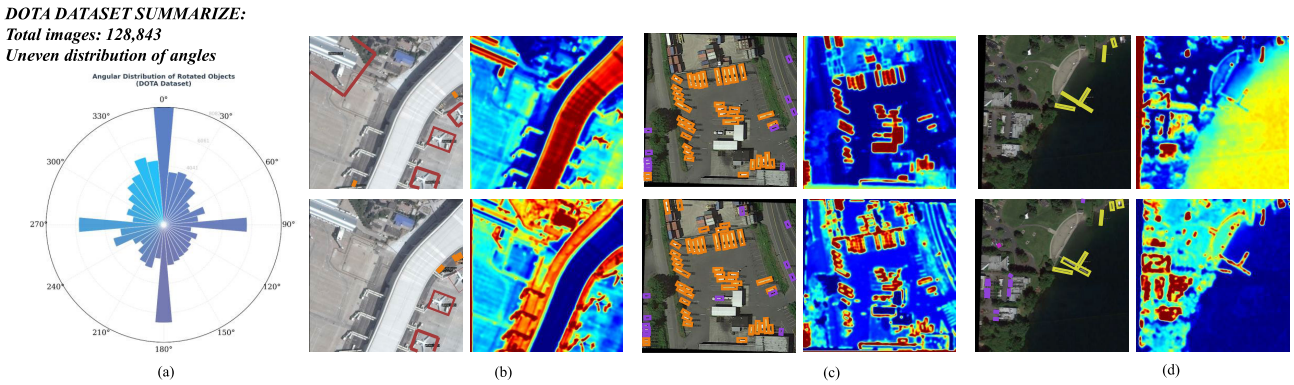


Fig. 1. Motivation and visual comparison of RIEMamba. (a) Angular distribution of oriented objects in the DOTA dataset; the total count shown in (a) denotes sliced image patches used for angular-statistics visualization rather than original DOTA scenes. (b)–(d) Detection results (odd columns) and corresponding feature heatmaps (even columns) for three representative challenges: (b) rotation, (c) edge ambiguity, and (d) scale variation. In each group, the top row shows the baseline and the bottom row shows RIEMamba.

case (c), the baseline detects many vehicles in the parking area but its heatmap spreads into the asphalt background and neighboring vehicles, which is consistent with boundary confusion and duplicate or inaccurate localization. RIEMamba keeps the responses tighter on the vehicle interiors and boundaries, reducing background activation around the parked rows. In the scale case (d), the baseline misses several small aircraft-like targets in the open area and shows weak responses near the water-land boundary, whereas RIEMamba detects more small instances and maintains clearer local responses while preserving contextual support from the surrounding scene. These examples indicate that RIEMamba improves rotation consistency, boundary focus, and scale-aware context modeling; the actual detection improvement is further supported by quantitative backbone comparisons, component ablations, and error analysis in Section IV. Mamba provides a promising backbone direction for this setting because its state-space sequence modeling captures long-range dependencies with linear complexity, making it suitable for high-resolution remote sensing images that contain sparse targets, dense object groups, and broad contextual layouts. Nevertheless, existing visual Mamba models mainly address generic sequence modeling and do not explicitly encode the geometric and structural priors required by oriented RSOD. Vanilla Mamba therefore alleviates the long-range context problem to some extent, but it still lacks dedicated mechanisms for rotation consistency, boundary preservation, and scale adaptation.

To address these remaining deficiencies, we revisit foundational visual priors in remote sensing and propose a novel backbone grounded in SO(2)-based [13] rotational equivariance. Specifically, we develop feature extractors that are robust to orientation uncertainty, where differential convolutions integrating Sobel operators and local binary patterns (LBPs) enhance edge-texture representation. Furthermore, a hierarchical Mamba scanning mechanism is introduced to aggregate multi-scale contextual gradients.

Building on these insights, we propose RIEMamba, a unified Mamba-based backbone that jointly addresses rotation sensitivity, edge ambiguity, and scale variation for oriented RSOD. Rather than treating these challenges independently,

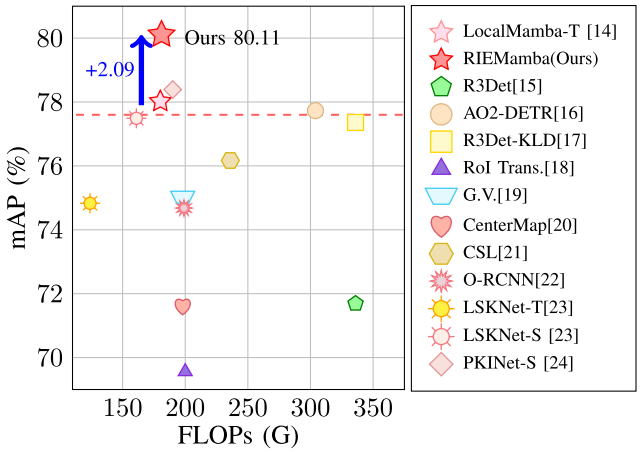


Fig. 2. Comprehensive evaluation of the detection performance and computational cost of different remote sensing object detectors on the single-scale DOTA-v1.0 benchmark.

RIEMamba couples rotation-aware feature extraction with adaptive multi-scale context aggregation within a single architecture, forming a cohesive learning paradigm that delivers geometrically stable, semantically expressive, and computationally efficient representations. As shown in Fig. 2, our model achieves superior performance-efficiency trade-offs compared to a wide range of state-of-the-art detectors, with clear gains in accuracy under comparable computational budgets.

The main contributions of this study are as follows.

1) We design RIEMamba, a Mamba-based backbone tailored to oriented RSOD. It preserves Mamba's efficient long-range sequence modeling while introducing rotation-consistent, edge-aware, and scale-adaptive priors required by complex aerial scenes.
2) We propose RIEPDC, a novel Rotation-Invariant Edge-aware Pixel Difference Convolution (RIEPDC), which integrates SO(2)-based kernel transformations with Sobel gradient and LBP operators. This hybrid design captures rotation-consistent structural cues and texture primitives, enabling the network to perceive

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edges and object boundaries under arbitrary orientations and varying illumination conditions.

3) We introduce variable multi-scale scanning (VMSS), a dual-branch VMSS strategy, which adaptively fuses information from heterogeneous receptive fields and dynamically allocates attention across object scales. This mechanism strengthens scale-aware context reasoning and enhances the detection of small and densely distributed targets that are prevalent in remote sensing images.
4) The proposed RIEMamba achieves state-of-the-art performance on multiple benchmarks including DOTA, HRSC2016, and DIOR-R, while maintaining a low computational footprint. The results demonstrate the strong generalization capability and efficiency of our method in complex aerial detection scenarios.

The remainder of this article is organized as follows. Section II reviews related works on rotational equivariant modeling, Mamba-based architectures, and RSOD. Section III details the proposed RIEMamba framework, including the design of the RIEPDC module and the VMSS multi-scale scanning strategy. Section IV presents the experimental results and comprehensive analyses on multiple benchmark datasets, validating the effectiveness and efficiency of the proposed method. Finally, Section V concludes this article and discusses potential directions for future research.

II. Related Works

A. Oriented RSOD

RSOD has largely followed general object detection paradigms [3], evolving into CNN-based and Transformer-based approaches. Among CNN-based methods, two-stage detectors [22], [25], [26] refine region proposals with orientation-aware modules. SCRDet enhances salient regions with attention and loss design, while Oriented R-CNN introduces angle-sensitive proposals. Single-stage methods [15], [27], [28], [29], [30] focus on direct angle regression and dense anchor alignment. R3Det employs multistep refinement, and RSDet++ improves angular stability. Transformer-based methods improve global modeling and spatial adaptability. GGHL [31] encodes orientation and shape information via 2-D Gaussian heatmaps to achieve stable label assignment and accurate angle regression. AO2-DETR [16], ARS-DETR [32], EMO2-DETR [33], and QETR [34] propose rotation-aware attention, deformable designs, or enhanced label assignment. LSKNet [23] uses large kernels for efficient context encoding, and PKINet [24] introduces keypoint-guided spatial refinement. Pixel-difference modeling has also been adapted to resource-efficient RSOD; HBiPiDiNet [35] combines binary convolution with local difference cues to preserve object contrast under aggressive network quantization. However, most existing methods neglect explicit orientation-consistent feature modeling, a key factor in remote sensing scenarios with arbitrary object orientations. They often depend on post-processing or angle regression, lacking structural priors in the backbone. Our approach differs from rotation-augmented training or angle regression methods by embedding SO(2)-equivariant priors directly into the convolution kernels, enabling more consistent orientation-aware representations.

B. Mamba Architecture

Thanks to its state-space formulation and efficient long-range dependency modeling [36], Mamba [37] has shown strong advantages over CNNs and Transformers in dense prediction tasks. Compared with CNNs, Mamba can propagate information over a broader spatial extent without relying solely on stacked local kernels; compared with self-attention-based Transformers, it models long image sequences with linear complexity rather than quadratic token interactions. These properties are attractive for remote sensing imagery, where high-resolution scenes require both global context reasoning and computationally feasible dense prediction. Recent works have introduced Mamba into object detection and segmentation pipelines using remote sensing datasets. RS-Mamba [38] improves semantic representation via dynamic aggregation. Pan-Mamba [39] adopts lightweight channel interaction for cross-modal fusion. Encoder-decoder structures such as SAMBA [40] and RS-SAMamba [41] enhance segmentation performance through semantic modeling, while omnidirectional scanning [42] aids context integration. However, these models largely follow general object detection paradigms and do not consider the unique demands of remote sensing, such as scale diversity and background clutter, let alone the challenges posed by rotated object detection. Oriented RSOD requires precise localization under arbitrary orientations, which is rarely addressed in existing Mamba-based frameworks. To fill this gap, we propose a rotation-aware Mamba architecture that integrates orientation-equivariant modeling and multi-scale adaptive fusion, tailored for the geometric complexity of aerial scenes.

C. General Rotation and Equivariance Modeling

In recent years, rotation and equivariance modeling has become an essential direction in visual representation learning, aiming to ensure that model responses remain invariant or transform equivalently when the input undergoes geometric rotation. Early studies mainly focused on constraining convolutional kernels to explicitly embed rotation invariance within convolution operations, thereby improving geometric generalization. For instance, local edge-based variants such as Sobel [43] and LBP [44] operators, together with sorted random projections, achieved robustness to orientation changes through feature reordering and local gradient encoding. Harmonic Networks [45] parameterized convolutional kernels as circular harmonic functions, thereby guaranteeing continuous rotation equivariance in theory. Subsequently, E(2)-Steerable CNNs [46] and SRE-Conv [47] introduced group constraints and symmetric ring structures into kernel design, jointly realizing rotation and reflection equivariance while maintaining compactness. These approaches established mechanism-assured rotation invariance through structural priors, reducing reliance on extensive rotational data augmentation. With the advance of group theory, research gradually shifted toward more general frameworks of equivariant modeling. Group Equivariant CNNs

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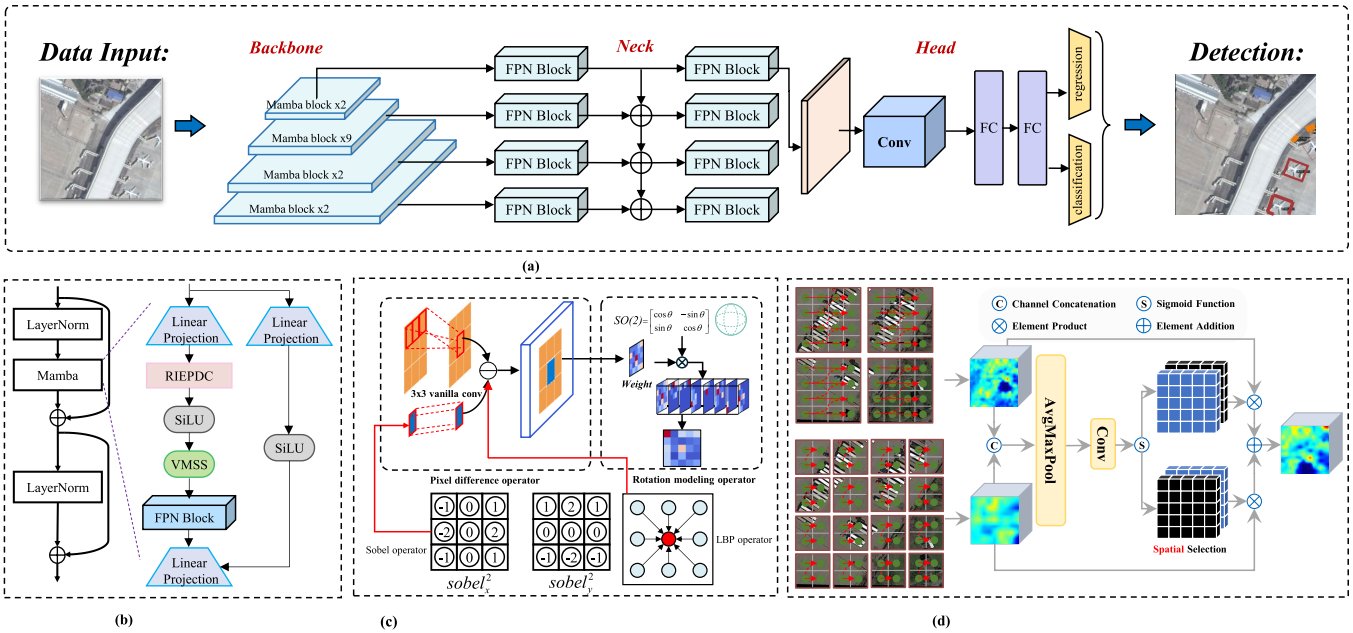


Fig. 3. Overview of the proposed RIEMamba architecture. (a) Overall backbone structure. (b) Mamba block incorporating RIEPDC and VMSS. (c) RIEPDC. (d) VMSS.

(G-CNNs) [48] were the first to define convolution operators on group spaces, naturally extending the translation group to the rotation groups SO(2) and SO(3). Steerable CNNs [49] then decomposed filters into steerable bases and feature types, allowing kernels to share parameters not only across spatial positions but also across orientations. Further developments such as LieConv [50] and Clifford-Steerable CNNs [51] generalized this idea to continuous Lie group settings, enabling consistent geometric representation across arbitrary transformations. In parallel, the Rotation Equivariant Vector Field Network (RotEqNet) [52] explicitly constructed vector fields to preserve dominant orientation information under rotation, inspiring later studies on directional field modeling and multitask applications. More recently, RIS-PiDiNet [53] jointly embeds rotation-group priors and structural symmetry constraints into pixel-difference convolution for RSOD, demonstrating the value of geometrically structured local operators. Overall, the evolution of general rotation and equivariance modeling has progressed from kernel-constraint mechanisms to group-representation learning and finally to structured deep integration, consistently striving to achieve stable and computationally efficient orientation modeling, especially vital for high-resolution remote sensing applications. In this article, rotation-consistent or SO(2)-equivariant features refer to feature-level responses that transform stably with in-plane rotations, whereas rotation-invariant detection refers to the final detection outputs being more stable under object rotation after feature aggregation and prediction.

III. Proposed Method

A. Overview of RIEMamba

Fig. 3 shows the overall architecture of RIEMamba. The backbone follows a four-stage hierarchical design with Mamba blocks arranged as 2, 9, 2, and 2 blocks from shallow to deep stages. The multi-scale feature maps are passed to an FPN neck and then to an oriented detection head for classification and bounding-box regression. As shown in Fig. 3(b), each Mamba block contains the original Mamba sequence modeling path together with RIEPDC and VMSS. Fig. 3(c) and (d) illustrate the internal structures of RIEPDC and VMSS, respectively. RIEMamba denotes the complete backbone throughout this article, while RIEPDC and VMSS refer to its two inserted modules.

B. Rotation Invariant and Edge Aware Pixel Difference Convolution

As illustrated in Fig. 3(c), this module is designed to enhance rotation-consistent edge representation by encoding orientation-aware contours and suppressing background interference. Difference-based operators have proven effective for efficient contrast and boundary representation in general salient object detection [54] and optical remote sensing salient object detection [55], [56]. It integrates SO(2)-based kernel transformations to capture continuous rotational symmetry, while incorporating local descriptors such as LBP and Sobel to reinforce edge-aware responses. By explicitly embedding this rotational structure, RIEPDC enables the model to distinguish object boundaries more consistently under arbitrary in-plane orientations in complex remote sensing scenes.

To address rotation-level variability in remote sensing scenes, we introduce a rotation-consistent convolution kernel transformation grounded in the Lie group SO(2), the group of planar rotations. Mathematically, the group SO(2) is defined as follows:

$$\mathrm{SO}(2) = \left\{ R(\alpha) = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix} : \alpha \in [0, 2\pi) \right\} \quad (1)$$

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which represents all continuous rotations in the 2-D Euclidean plane. Given a convolutional kernel $w \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}} \times k \times k}$, we define its SO(2)-induced rotated form as follows:

$$\widetilde{w}_\alpha(u) = w(R(\bar{\alpha})u), \quad \bar{\alpha} = -\alpha, \quad u \in \mathbb{R}^2 \tag{2}$$

where w denotes the original convolution kernel, $\widetilde{w}_\alpha$ is the rotated version of the kernel, $u \in \mathbb{R}^2$ represents a spatial coordinate in the kernel domain, and $R(\bar{\alpha}) \in \text{SO}(2)$ is the rotation matrix with the negative angle $\bar{\alpha} = -\alpha$, which maps the rotated coordinates back to the original kernel domain via backward warping. By applying SO(2) kernel transformations, the network encodes continuous rotational priors, promoting stable responses for rotated objects without relying solely on their canonical poses.

To enhance robustness under arbitrary rotations, edge ambiguities, and scale changes, we propose a RIEPDC that computes features from local intensity differences rather than raw intensities. The overall formulation of RIEPDC is defined as follows:

$$\begin{aligned} y(p_0) = & \underbrace{\sum_{p_n \in \mathcal{R}} w(p_n)\, x(p_0 + p_n)}_{\text{vanilla conv}} - \theta\, x(p_0) \\ & \underbrace{\sum_{p_n \in \mathcal{R}} w(p_n)\, \left\| \nabla x(p_0 + p_n) \right\|}_{\text{pixel diff}} \\ & + \beta \underbrace{\sum_{p_n \in \mathcal{R}} (\widetilde{w}_\alpha(p_n) - w(p_n))\, x(p_0 + p_n)}_{\text{rotation modeling}} \end{aligned} \tag{3}$$

where p_0 denotes the central pixel and $p_n \in \mathcal{R}$ enumerates neighboring positions within the receptive field. The first term is a standard convolution using kernel w. The second term introduces edge-aware pixel difference, where the response at p_0 is modulated by the Sobel gradient magnitude $\|\nabla x\| = (\text{sobel}_x^2 + \text{sobel}_y^2)^{1/2}$, emphasizing structural boundaries. The third term adds a rotation-aware residual, computed from the difference between the rotated kernel $\widetilde{w}_\alpha$ and the original kernel w, capturing local orientation consistency. The scalar θ controls the edge-aware suppression strength, and β adjusts the residual orientation consistency. This formulation jointly improves robustness to blurry edges and orientation variation, which are prevalent in remote sensing imagery.

C. Variable Multi-Scale Scanning

We propose the VMSS technique, as shown in Fig. 3(d). This technique is designed to enhance the model's adaptability to objects of diverse sizes in high-resolution remote sensing imagery. Unlike fixed receptive field designs, VMSS performs scanning at multiple scales with dynamically adjustable strides and receptive fields, thereby allowing the network to capture both fine-grained local structures and broader contextual information. Specifically, the input features are projected into two branches, and selective scanning is performed in different directions to obtain feature maps

$$\mathcal{H}_1 = \text{Scan}(\mathcal{U}_1), \quad \mathcal{H}_2 = \text{Scan}(\mathcal{U}_2) \tag{4}$$

where $\mathcal{U}_1$ and $\mathcal{U}_2$ denote the projected feature maps of the two branches.

To aggregate complementary information, average pooling and max pooling are applied to the concatenated feature maps

$$\mathcal{G}_{\text{avg}} = \text{Avg}(Cat(\mathcal{H}_1, \mathcal{H}_2)), \quad \mathcal{G}_{\text{max}} = \text{Max}(Cat(\mathcal{H}_1, \mathcal{H}_2)) \tag{5}$$

where $Avg(\cdot)$ and $Max(\cdot)$ represent the average and maximum pooled descriptors, respectively.

A gating vector is then generated by applying a Sigmoid activation

$$\mathcal{G} = \sigma\left(\text{Cat}\left(\mathcal{G}_{\text{avg}}, \mathcal{G}_{\text{max}}\right)\right) \tag{6}$$

where $\sigma(\cdot)$ denotes the Sigmoid function. To assign branch-specific weights, $\mathcal{G}$ is split along the channel dimension as follows:

$$\mathcal{G}_1, \mathcal{G}_2 = \text{Split}(\mathcal{G}), \quad \mathcal{G}_i \in \mathbb{R}^C,\ i = 1, 2 \tag{7}$$

where $\mathcal{G}_1$ and $\mathcal{G}_2$ correspond to the adaptive weighting coefficients of the two scanning branches. Finally, the scanning outputs are fused as follows:

$$\mathbf{S} = \mathcal{G}_1 \odot \mathcal{H}_1 + \mathcal{G}_2 \odot \mathcal{H}_2 \tag{8}$$

where $\odot$ represents element-wise multiplication.

This technique focuses on adaptive receptive field aggregation through VMSS, which enhances the model's ability to capture scale-aware context and improves robustness to object size variation in remote sensing scenes.

IV. Experiments

A. Datasets Description

DOTA-v1.0 [57] comprises 2806 remote sensing images, encompassing 188 282 instances across 15 categories.

Similarly, HRSC2016 [58] is a popular dataset for rotated ship detection. It contains 1061 images and 2976 ship instances, with sizes ranging from 300 × 300 to 1500 × 900 pixels.

DIOR-R [59], an aerial image dataset derived from DIOR [2] with Oriented Bounding Box (OBB) annotations, includes 23 463 images, 20 categories, and 190 288 instances.

B. Evaluation Metrics

We adopt mean Average Precision (mAP) as the primary evaluation metric across all benchmarks. For a given category, the Average Precision (AP) is computed as the area under the precision-recall curve

$$\text{AP} = \int_0^1 p(r)\, dr \tag{9}$$

where $p(r)$ denotes the precision at recall level r. The mean AP mAP is then obtained by averaging AP over all C categories

$$\text{mAP} = \frac{1}{C} \sum_{i=1}^{C} \text{AP}_i. \tag{10}$$

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TABLE I

EFFECTIVENESS OF RIEMAMBA BACKBONE ON VARIOUS RSOD FRAMEWORKS. PARAMETERS AND FLOPS ARE BACKBONE-ONLY. TRAINING TIME (T) PER EPOCH AND PEAK GPU MEMORY (MEM) ARE MEASURED ON 4× A800-80 GB WITH BATCH SIZE 8 AND 1024^2 INPUT. THE RIEMAMBA BACKBONE IS PRETRAINED ON IMAGENET FOR 300 EPOCHS

Method	Backbone	B.#P	B.FLOPs	mAP	FPS	T(h)	Mem(G)
O-RCNN [22]	ResNet-50 [80]	23.3M	86.1G	77.49	15.6	3.1	24.3
	Swin-T [81]	27.5M	94.0G	78.36	13.2	3.7	30.5
	LSKNet-T [23]	4.3M	18.8G	77.85	14.8	3.3	18.7
	LocalMamba-T [14]	34.3M	105.21G	78.02	11.5	4.2	38.4
	RIEMamba	37.69M	106.17G	**80.11**	11.1	4.4	41.6
RoI Trans. [18]	ResNet-50 [80]	23.3M	86.1G	73.61	14.3	3.4	26.1
	Swin-T [81]	27.5M	94.0G	76.42	12.1	4.0	32.3
	LSKNet-T [23]	4.3M	18.8G	75.89	13.5	3.6	20.8
	LocalMamba-T [14]	34.3M	105.21G	77.08	10.4	4.7	40.5
	RIEMamba	37.69M	106.17G	**79.73**	10.1	4.8	43.7
S^2ANet [27]	ResNet-50 [80]	23.3M	86.1G	77.42	16.8	2.9	22.8
	Swin-T [81]	27.5M	94.0G	78.07	14.2	3.4	28.7
	LSKNet-T [23]	4.3M	18.8G	77.73	15.9	3.1	17.4
	LocalMamba-T [14]	34.3M	105.21G	78.35	12.3	3.9	36.5
	RIEMamba	37.69M	106.17G	**79.12**	11.9	4.1	39.8
R3Det [15]	ResNet-50 [80]	23.3M	86.1G	76.47	14.9	3.3	25.4
	Swin-T [81]	27.5M	94.0G	77.25	12.6	3.9	31.7
	LSKNet-T [23]	4.3M	18.8G	76.91	14.1	3.5	19.8
	LocalMamba-T [14]	34.3M	105.21G	77.68	10.8	4.5	39.3
	RIEMamba	37.69M	106.17G	**78.63**	10.5	4.6	42.5
YOLO [69]	CSPNeXt-l [69]	52.3M	80.2G	75.14	19.4	2.5	21.4
	Swin-T [81]	27.5M	94.0G	75.68	14.7	3.3	27.9
	LSKNet-T [23]	4.3M	18.8G	75.31	16.2	3.0	16.8
	LocalMamba-T [14]	34.3M	105.21G	76.08	12.7	3.8	35.7
	RIEMamba	37.69M	106.17G	**76.62**	12.3	4.0	38.9
ARS-DETR [32]	ResNet-50 [80]	23.3M	86.1G	70.06	10.2	4.8	31.8
	Swin-T [81]	27.5M	94.0G	71.53	8.7	5.6	37.6
	LSKNet-T [23]	4.3M	18.8G	70.92	9.6	5.1	26.4
	LocalMamba-T [14]	34.3M	105.21G	71.86	7.4	6.5	45.7
	RIEMamba	37.69M	106.17G	**73.32**	7.2	6.7	48.3

A predicted OBB is considered a true positive if its Intersection over Union (IoU) with a ground-truth box exceeds a predefined threshold τ

$$\mathrm{IoU} = \frac{\left|B_{\mathrm{pred}} \cap B_{\mathrm{gt}}\right|}{\left|B_{\mathrm{pred}} \cup B_{\mathrm{gt}}\right|} \tag{11}$$

where B_{pred} and B_{gt} denote the predicted and ground-truth oriented bounding boxes, respectively. For DOTA-v1.0 and DIOR-R, mAP is computed under the PASCAL VOC 2007 protocol with $\tau = 0.5$. For HRSC2016, we report results under both VOC 2007 [mAP(07)] and VOC 2012 [mAP(12)] metrics to facilitate comprehensive comparison with prior work.

To evaluate model efficiency, we report the number of parameters (#P) and floating-point operations (FLOPs). In Table I, we further report inference speed (FPS), training time per epoch (T), and peak GPU memory consumption (Mem) under a unified hardware protocol. All mAP values are reported to two decimal places, whereas #P and FLOPs retain the precision available in the corresponding source publications to avoid implying unsupported measurement accuracy. In all tables, **bold** values indicate the best result, and underlined values indicate the second-best result.

C. Implementation Details

Our method is implemented on MMRotate with PyTorch 1.12.1. Backbones are pre-trained on ImageNet-1K via MMClassification and fine-tuned for remote sensing detection. Pre-training uses Mamba variants (e.g., VMamba, LocalMamba) on 8× L20 GPUs (48 GB) with batch size 256, learning rate 5×10^{-5}, AdamW optimizer, and 100 epochs. For oriented object detection, we adopt the same Oriented RCNN framework with LSKNet, LocalMamba-T, and RIEMamba backbones, and train them on 4× A800-80GB GPUs (batch size 8, lr 5×10^{-5}, 12 epochs). LocalMamba-T is obtained by replacing the RIEMamba backbone with the LocalMamba-T backbone while keeping the same detector, FPN neck, input size, optimizer, and training schedule. Backbone-only parameters and FLOPs are separately reported in Table I, whereas Tables II–IV report detector-level costs. We evaluate on DOTA-v1.0 (single-scale and multi-scale: resize $\{0.5, 1.0, 1.5\}$, crop 1024^2 with 500 overlap), HRSC2016 (resize longer side to 800), and DIOR-R. All results are reported on standard train/val/test splits following common practice. To substantiate computational efficiency, all efficiency-related metrics are reported under a unified hardware protocol (4× A800-80 GB, batch size 8, input 1024^2): training time per epoch, inference FPS, and peak GPU memory are listed alongside parameters and FLOPs in Table I. Note that #P and FLOPs in Table II are identical for single-scale and multi-scale settings because they describe the same detector architecture under different test-scale protocols, and are therefore independent of the multi-scale testing strategy.

D. Comparison With State-of-the-Art Methods

We compared our RIEMamba model against several state-of-the-art methods on the DOTA-v1.0 dataset, which includes several implementations of the Mamba-based backbone, as shown in Table II. Notably, our RIEMamba models achieved state-of-the-art detection mAP.

1) Results on DOTA-v1.0: As shown in Table II, our RIEMamba achieves 80.11% (single-scale) and 82.49% (multi-scale), surpassing LocalMamba-T by 2.09% and 1.57%, respectively. Category-wise, our method shows clear gains on rotation-sensitive and elongated objects such as BR (+ 5.91%), SH (+ 0.56%), HA (+ 5.43%), and HC (+ 10.97%), benefiting from the rotation-invariant pixel-difference design and multi-scale scanning. Overall, RIEMamba excels on challenging categories while maintaining competitive results across the dataset.

2) Results on HRSC2016: As shown in Table III, RIEMamba achieves 90.60% mAP under VOC 2007% and 98.62% mAP under VOC 2012. It obtains the best VOC 2012 result and the second-best VOC 2007 result, only 0.05% behind PKINet-S, while maintaining moderate parameters and computational cost.

3) Results on DIOR-R: We evaluate the performance of our proposed RIEMamba on the DIOR-R dataset, and compare it with ten representative state-of-the-art methods, as summarized in Table IV. RIEMamba achieves 68.55% mAP, outperforming all compared methods. Compared with strong convolutional baselines such as PKINet-S (67.03%) and LSKNet-S (65.90%), RIEMamba shows clear advantages, validating the effectiveness of integrating long-range dependency modeling with orientation-consistent local representation. Furthermore, methods such as ARS-DETR and DCFL, which employ DETR-style detection heads or dynamic context fil-

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TABLE II

COMPARISON WITH STATE-OF-THE-ART METHODS ON THE DOTA-V1.0 DATASET (SINGLE-SCALE AND MULTI-SCALE). PRE.: IN = IMAGENET, CO = COCO, MA = MILLIONAID

Method	Pre.	mAP↑	#P↓	FLOPs↓	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC
Single-Scale																			
R3Det [15]	IN	71.69	41.9M	336G	89.54	81.99	48.46	62.52	70.48	74.29	77.54	90.80	81.39	83.54	61.97	59.82	65.44	67.46	60.05
CFA [60]	IN	75.05	–	–	89.26	81.72	51.81	67.17	79.99	78.25	84.46	90.77	82.40	85.54	54.86	67.75	73.04	70.24	64.96
SASM [61]	IN	74.92	–	–	86.42	78.97	52.47	69.84	77.30	75.99	86.72	90.89	82.63	85.66	60.13	68.25	73.98	72.22	62.37
AO2-DETR [16]	IN	77.73	74.3M	304G	89.27	84.97	56.67	74.89	78.87	82.73	87.35	90.50	84.68	85.41	61.97	69.96	74.68	72.39	71.62
S2ANet [27]	IN	76.11	–	–	88.70	81.41	54.28	69.75	78.04	80.54	88.04	90.69	84.75	86.22	65.03	65.81	76.16	73.37	58.86
R3Det-KLD [17]	IN	77.36	41.9M	336G	89.90	84.17	55.80	69.35	78.72	84.08	87.00	89.75	84.32	85.73	64.74	61.80	76.62	**78.49**	70.89
SCRDet [26]	IN	72.61	–	–	89.98	80.65	52.09	68.36	68.36	60.32	72.41	90.85	87.94	**86.86**	65.02	66.68	66.25	68.24	65.21
RoI Trans [18]	IN	69.56	55.1M	200G	88.64	78.52	43.44	75.92	68.81	73.68	83.59	90.74	77.27	81.46	58.39	53.54	62.83	58.93	47.67
G.V. [19]	IN	75.02	41.1M	198G	89.64	85.00	52.26	77.34	73.01	73.14	86.82	90.74	79.02	86.81	59.55	70.91	72.94	70.86	57.32
CenterMap [20]	IN	71.59	41.1M	198G	89.02	80.56	49.41	61.98	77.99	74.19	83.74	89.44	78.01	83.52	47.64	65.93	63.68	67.07	61.59
CSL [21]	IN	76.17	37.4M	236G	**90.25**	85.53	54.64	75.31	70.44	73.51	77.62	90.84	86.15	86.69	**69.60**	68.04	73.83	71.10	68.93
ReDet [62]	IN	76.25	–	–	88.79	82.64	53.97	74.00	78.13	84.06	88.04	90.89	87.78	85.75	61.76	60.39	75.96	68.07	63.59
DODet [63]	IN	75.89	–	–	89.61	83.10	51.43	72.02	79.16	81.99	87.71	90.89	86.53	84.56	62.21	65.38	71.98	70.79	61.93
AOPG [59]	IN	75.39	–	–	89.14	82.74	51.87	69.28	77.65	82.42	88.08	90.89	86.26	85.13	60.60	66.30	74.05	67.76	58.77
GGHL [31]	IN	76.95	–	–	89.74	85.63	44.50	77.48	76.72	80.45	86.16	90.83	**88.18**	86.25	67.07	69.40	73.38	68.45	70.14
PCI [30]	IN	76.32	–	–	89.89	**86.63**	53.38	77.87	76.78	79.92	87.36	90.86	84.65	85.35	65.69	61.25	74.64	70.39	60.76
O-RCNN [22]	IN	74.68	41.1M	199G	89.71	79.65	49.80	70.59	77.85	76.95	87.45	90.89	85.83	83.95	59.28	65.56	66.35	70.63	65.82
LSKNet-T [23]	IN	74.83	**21.0M**	**124G**	89.88	75.08	51.01	72.44	78.23	82.82	88.04	90.90	85.97	84.13	60.05	63.62	67.51	69.11	63.64
LSKNet-S [23]	IN	77.49	31.0M	161G	89.66	85.52	57.72	75.70	74.95	78.69	88.24	90.88	86.79	86.38	66.92	63.77	77.77	74.47	64.82
PSC [64]	IN	78.10	–	–	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
PKINet-S [24]	IN	78.39	30.8M	190G	89.72	84.20	55.81	77.63	**80.25**	84.45	88.12	90.88	87.57	86.07	66.86	70.23	77.47	73.62	62.94
LocalMamba-T [14]	IN	78.02	42.0M	180.11G	89.47	83.70	54.40	75.87	79.90	**84.93**	87.94	90.87	87.07	85.15	61.96	**72.62**	77.26	71.85	67.32
RIEMamba	IN	**80.11**	45.39M	181.07G	89.20	84.49	**60.31**	**78.13**	79.12	83.60	**88.50**	**90.94**	85.13	86.81	65.05	70.32	**82.69**	77.06	**78.29**
Multi-Scale																			
CSL [65]	IN	76.17	37.4M	236G	90.25	85.53	54.64	75.31	70.44	73.51	77.62	90.84	86.15	86.69	69.60	68.04	73.83	71.10	68.93
CFA [60]	IN	76.67	–	–	89.08	83.20	54.37	66.87	81.23	80.96	87.17	90.21	84.32	86.09	52.34	69.94	75.52	80.76	67.96
DAFNet [66]	IN	76.95	–	–	89.40	86.27	53.70	60.51	82.04	81.17	88.66	90.37	83.81	87.27	53.93	69.38	75.61	81.26	70.86
DODet [67]	IN	80.62	–	–	89.96	85.52	58.01	81.22	78.71	85.46	88.59	90.89	87.12	87.80	70.50	71.54	82.06	77.43	74.47
AOPG [59]	IN	80.66	–	–	89.88	85.57	60.90	81.51	78.70	85.29	88.85	90.89	87.60	87.65	71.66	68.69	82.31	77.32	73.10
KFloU [68]	IN	80.93	58.8M	206G	89.44	84.41	**62.22**	82.51	80.10	86.07	88.68	90.90	87.32	**88.38**	72.80	71.95	78.96	74.95	75.27
RTMDet-R [69]	CO	81.33	52.3M	205G	88.01	86.17	58.54	82.44	81.30	84.82	88.71	90.89	**88.77**	87.37	71.96	71.18	81.23	81.40	77.13
RVSA [70]	MA	81.24	114.4M	414G	88.97	85.76	61.46	81.27	79.98	85.31	88.30	90.84	85.06	87.50	66.77	73.11	**84.75**	81.88	77.58
PCI [30]	IN	80.15	–	–	89.26	86.24	59.42	80.56	77.24	**87.45**	88.34	90.45	88.45	87.42	**74.23**	70.78	78.24	76.56	67.62
O-RCNN [22]	IN	80.87	41.1M	199G	89.84	85.43	61.09	79.82	79.71	85.35	88.82	90.88	86.68	87.73	72.21	70.80	82.42	78.18	74.11
LSKNet-T [23]	IN	81.37	**21.0M**	**124G**	89.14	84.90	61.78	**83.50**	81.54	85.87	88.64	90.89	88.02	87.31	71.55	70.74	78.66	79.81	**78.16**
LocalMamba-T [14]	IN	80.92	42.0M	180.11G	89.25	84.62	58.86	82.41	79.22	85.80	88.32	90.86	87.10	86.22	69.38	72.82	79.47	82.38	77.17
RIEMamba	IN	**82.49**	45.39M	181.07G	**90.67**	**86.44**	60.13	83.44	**82.82**	86.82	**89.33**	**90.99**	88.32	87.12	72.30	**74.14**	82.23	**84.13**	77.12

Note. Class abbreviations: PL = plane, BD = baseball diamond, BR = bridge, GTF = ground track field, SV = small vehicle, LV = large vehicle, SH = ship, TC = tennis court, BC = basketball court, ST = storage tank, SBF = soccer-ball field, RA = roundabout, HA = harbor, SP = swimming pool, and HC = helicopter.

tering, also fall behind RIEMamba in final performance. Notably, RIEMamba achieves a 2.96% mAP improvement over LocalMamba-T (65.59%), which demonstrates that the proposed RIEPDC and VMSS modules effectively enhance both the edge-aware feature extraction and multi-scale contextual aggregation capabilities for oriented object detection in diverse remote sensing scenes.

E. Ablation Studies

In this section, we report the results of ablation studies conducted on the single scale of DOTA-v1.0 test set to investigate the effectiveness of each method proposed in this article.

1) Effectiveness of Individual Components: As shown in Table V, to investigate the effectiveness of each method proposed in this article, we conducted ablation studies. The experiments demonstrate that the two proposed components in our RIEMamba, namely RIEPDC and VMSS, are both effective. RIEPDC does not increase the number of parameters and adds only negligible computational overhead (+ 0.25G FLOPs), yet brings notable performance improvements. VMSS, with only marginal increases in computational cost and parameters,

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TABLE III

COMPARISON WITH STATE-OF-THE-ART METHODS ON THE HRSC2016 DATASET

Method	mAP(07)	mAP(12)	#P	FLOPs
DRN [71]	–	92.70	–	–
CenterMap [20]	–	92.80	41.1M	198G
RoI Trans. [18]	86.20	–	55.1M	200G
G.V. [19]	88.20	–	41.1M	198G
R3Det [15]	89.26	96.01	41.9M	198G
DAL [72]	89.77	–	36.4M	216G
GWD [73]	89.85	97.37	47.4M	456G
S2ANet [27]	90.17	95.01	38.6M	198G
AOPG [59]	90.34	96.22	–	–
ReDet [62]	90.46	97.63	31.6M	–
O–RCNN [22]	90.50	97.60	41.1M	199G
LSKNet–T [23]	90.54	98.13	**21.0M**	**124G**
PKINet–S [24]	**90.65**	98.54	30.5M	159G
LocalMamba–T [14]	90.30	96.61	42.0M	180.11G
RIEMamba	90.60	**98.62**	45.39M	181.07G

TABLE IV

COMPARISON WITH STATE-OF-THE-ART METHODS ON THE DIOR-R DATASET. PRE.: IN = IMAGENET

Method	Pre.	#P	FLOPs	mAP↑
RetinaNet-O [74]	IN	–	–	57.55
Faster RCNN-O [75]	IN	41.1M	198G	59.54
TIOE-Det [76]	IN	41.1M	198G	61.98
O-RCNN [22]	IN	41.1M	199G	64.30
ARS-DETR [32]	IN	41.1M	198G	66.12
O-RepPoints [77]	IN	36.6M	–	66.71
DCFL [78]	IN	–	–	66.80
LSKNet-S [23]	IN	31.0M	**161G**	65.90
PKINet-S [24]	IN	**30.8M**	190G	67.03
LocalMamba-T [14]	IN	42.0M	180.11G	65.59
RIEMamba	IN	45.39M	181.07G	**68.55**

TABLE V

EFFECTIVENESS OF INDIVIDUAL COMPONENTS ON DOTA-V1.0

RIEPDC	VMSS	#P↓	FLOPs↓	mAP↑
		42.0M	180.11G	78.02
✓		42.0M	180.36G (+0.25)	79.38 (+1.36)
	✓	45.39M (+3.39)	180.82G (+0.71)	78.71 (+0.69)
✓	✓	45.39M (+3.39)	181.07G (+0.96)	**80.11** (+2.09)

TABLE VI

MAP ON DIFFERENT WINDOW SETTINGS ON DOTA-V1.0

Window	#P	FLOPs	mAP↑
$[2, S]$	45.39M	181.07G	**80.11**
$[2, 7]$	45.39M	181.07G	79.78
$[5, 7]$	45.39M	181.07G	79.70
$[24, 7]$	45.39M	181.07G	79.47
$[24, 2]$	45.39M	181.07G	79.52
$[24, S]$	45.39M	181.07G	79.65
$[S]$	45.39M	181.07G	79.56

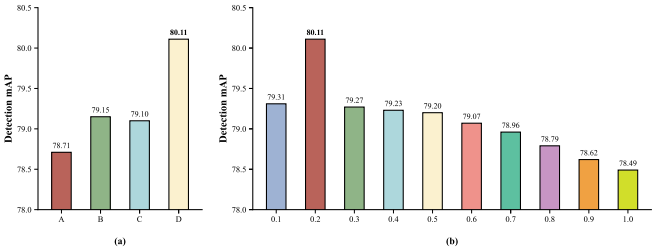


Fig. 4. (a) Ablation of convolutional variants within the RIEMamba framework on DOTA-v1.0, with the rest of the architecture kept unchanged. A denotes RIEMamba with vanilla convolution in place of RIEPDC, B denotes RIEMamba with SO(2)-based rotation modeling, C denotes RIEMamba with LBP+Sobel edge operators, and D denotes full RIEMamba with the complete RIEPDC combining SO(2), LBP, and Sobel. (b) mAP variation under different values of the hyperparameter θ, which controls the relative weighting of the differential term. These detection results complement the heatmap visualization in Fig. 1 by providing quantitative evidence that the proposed feature representation improves detection accuracy rather than only producing visually concentrated activations.

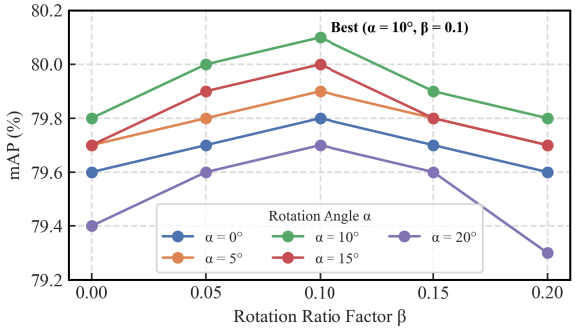


Fig. 5. Influence of different rotation angles and ratio factors on the accuracy of object detection.

also leads to a clear performance gain. The reasons are as follows.

1) RIEPDC combines the LBP operator and the Sobel operator to fully extract edge features from remote sensing images, laying the groundwork for the subsequent selective scanning mechanism.
2) VMSS aggregates features selectively scanned under two different window sizes, thereby enhancing multi-scale feature representation and improving detection accuracy. These results indicate that RIEPDC and VMSS address complementary aspects of oriented RSOD: RIEPDC strengthens rotation-consistent boundary representation, while VMSS improves scale-context balance; their combination yields the largest gain.

2) Impact of Convolutional Variants in RIEPDC: Fig. 4(a) evaluates four convolutional variants within a single RIEMamba setting. For all variants, the detector and the remaining RIEMamba components, including VMSS, are kept unchanged. Variant A replaces RIEPDC with vanilla convolution and obtains 78.71% mAP. Variant B introduces SO(2)-based rotation modeling and improves the result to 79.15%, while Variant C uses the LBP+Sobel edge operators and achieves 79.10%. Variant D is the complete RIEPDC with SO(2) LBP, and Sobel, reaching 80.11% mAP. These

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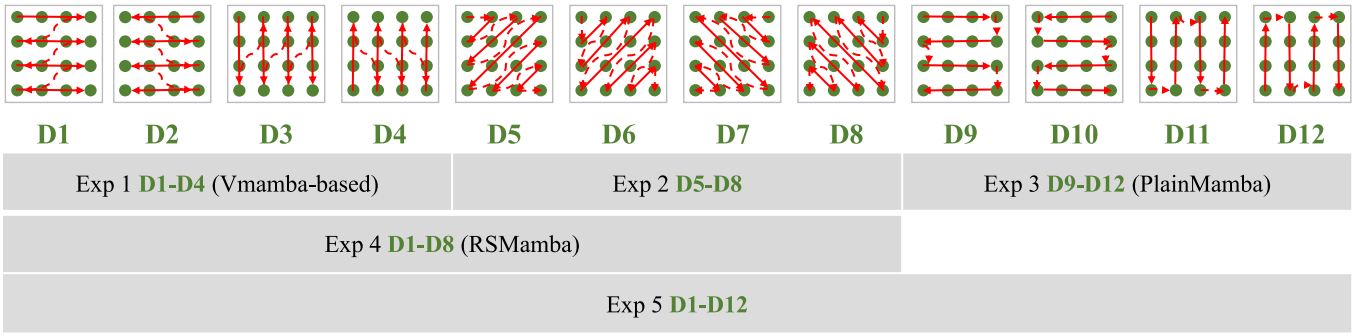


Fig. 6. Illustration of twelve spatial scanning variants (D1–D12) used to explore directional encoding in sequence modeling. Exp 1 (D1–D4) corresponds to VMamba-style horizontal and vertical patterns; Exp 2 (D5–D8) incorporates diagonal traversals to enhance spatial interaction; Exp 3 (D9–D12) represents PlainMamba-style row/column scans. Exp 4 combines D1–D8 for RS-Mamba, while Exp 5 utilizes all variants (D1–D12) for comprehensive spatial modeling.

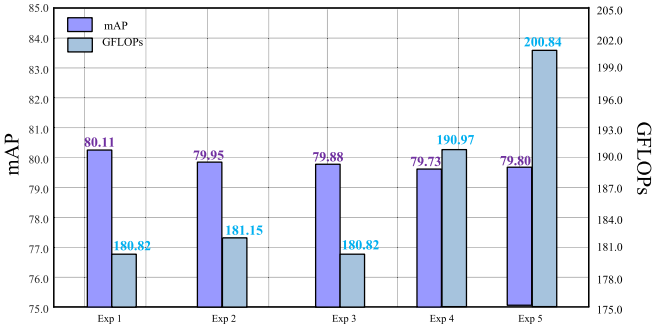


Fig. 7. Twelve spatial scanning patterns (D1–D12) grouped into five variants in Fig. 6 for ablation on RIEMamba, showing the impact of directional encoding on mAP and GFLOPs.

results show that rotation modeling and edge-aware difference operators are both beneficial, and their combination gives the strongest RIEMamba representation.

3) Impact of θ of RIEPDC: θ is used to balance the visual semantic information provided by the convolution and the gradient information provided by traditional operators. Broadly speaking, it modulates the focus between the basic attributes of the image's main subject, such as color and texture, and the attention to the object's edge features. An excessively high θ may lead to an overemphasis on edge features while neglecting the core attributes of the object. Conversely, θ that is too low may reduce the attention to edge features, compromising the discriminative ability of remote sensing object edges. The experimental data in Fig. 4(b) indicate that the optimal value for θ is 0.2.

4) Impact of Rotation Angle and Rotation Ratio Factor for RIEPDC: To improve robustness against variations in object orientation, we introduce angular perturbation during training. This strategy improves orientation robustness, but requires careful calibration: strong perturbations may lead to semantic drift due to misaligned features, whereas weak perturbations yield limited improvement. Empirical results in Fig. 5 demonstrate that setting rotation angle $\alpha = 10$ and $\beta = 0.1$ leads to optimal accuracy.

5) Impact of Window Size for VMSS: The selection of window sizes for the two branches is a critical decision in the design of the VMSS module. Larger windows capture more contextual information, while smaller windows focus more on local features of the object. The optimal combination of these two window sizes is a key determinant of the model's representational capability. These results verify that adaptive dual-scale scanning (2,S) achieves the optimal balance between local detail and global context. The chosen dual-scale setup was empirically optimal; future work could extend VMSS to a learnable dynamic-scale variant via attention or reinforcement search.

6) Impact of Different Scanning Directions: Scanning direction, though often overlooked in sequence modeling, critically affects spatial dependency learning in remote sensing imagery. In this ablation study, we evaluate twelve scanning variants grouped into five configurations in Fig. 6, ranging from horizontal and vertical to diagonal and multi-directional strategies within the Mamba framework. Results on RIEMamba in Fig. 7 show that adding more directional scan paths does not necessarily improve accuracy and can increase computational cost. Exp 1, which uses D1–D4 VMamba-style horizontal and vertical scanning [79], achieves the best trade-off between detection accuracy and computational cost among all evaluated variants. Its 181.07G FLOPs value is consistent with the complete RIEMamba cost reported in Tables II, V, and VI.

7) Backbone Generality Across Detectors: Table I keeps the detector, neck, input scale, and training setting fixed while replacing only the backbone, so the comparison isolates the effect of backbone representation. LocalMamba-T is obtained in the same way by replacing the RIEMamba backbone with LocalMamba-T. RIEMamba improves mAP in proposal-based pipelines such as O-RCNN and RoI Transformer and in dense prediction pipelines such as YOLO and ARS-DETR, indicating that the proposed representation is not tied to a specific detection head. The FPS, training-time, and memory columns report detector-level computational cost, whereas B.#P and B.FLOPs are backbone-only statistics.

F. Qualitative Analysis

We further examine RIEMamba through detection heatmaps, angular robustness, scene-level visual comparisons, TIDE error decomposition, and t-SNE feature distributions.

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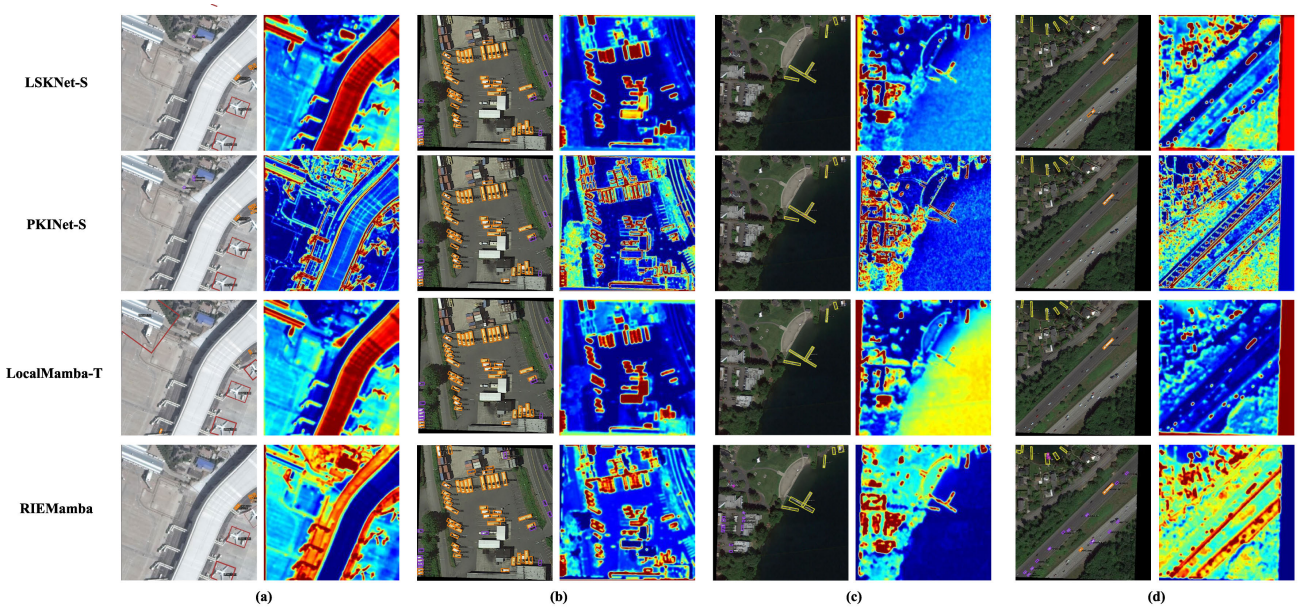


Fig. 8. Detection results and heatmap visualization on DOTA-v1.0. Odd columns show detection results and even columns show feature heatmaps for four representative scenarios: (a) typical scene, (b) scale variation, (c) boundary ambiguity, and (d) diverse object orientations.

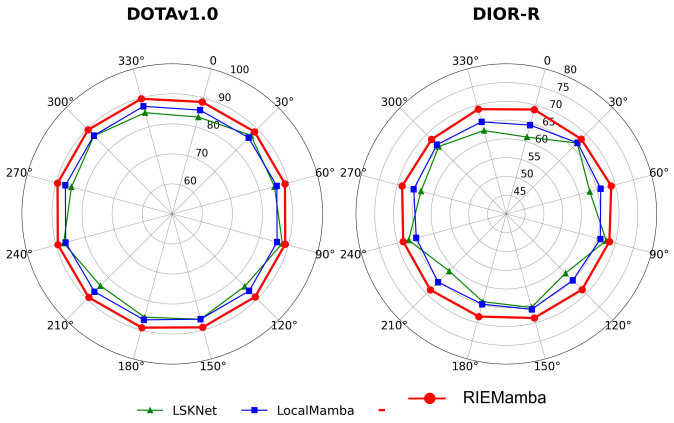


Fig. 9. Rotation robustness on DOTA-v1.0 val and DIOR-R test across multiple angles. In the legend, LSKNet and LocalMamba denote the LSKNet-T and LocalMamba-T variants used in the comparison.

These visual analyses are used as complementary evidence to the quantitative results rather than as standalone proof.

1) Detection and Heatmap Visualization: Fig. 8 compares detection outputs and heatmaps across typical, scale-varying, boundary-ambiguous, and diverse-orientation scenes. Baseline responses are often diffused over background regions or incomplete on object regions, which corresponds to false positives, missed detections, and inaccurate localization. In contrast, RIEMamba produces more object-centered activations and cleaner oriented boxes. Fig. 9 further shows that RIEMamba maintains more stable performance across rotation angles on DOTA-v1.0 and DIOR-R than LSKNet-T and LocalMamba-T, supporting its rotation-consistent representation.

2) Scene-Level Comparison and Error Analysis: Fig. 10 shows representative dense parking, recreational, transport-hub, and sports-ground scenes. Compared with LSKNet-S, RIEMamba reduces the missed detections, duplicate boxes, and false detections marked in the zoomed regions, especially for small and densely arranged objects. The TIDE analysis in Fig. 11 provides error-level evidence: RIEMamba reduces several major error sources, especially classification (Cls), background confusion (Bkg), missed ground truth (Miss), and FP/FN counts, while keeping localization-related errors within a competitive range.

A closer inspection of Fig. 8 further reveals how the proposed representation changes the failure patterns of the detector. In the typical scene in Fig. 8(a), the baseline heatmaps tend to respond not only to the target regions but also to strong background structures such as roads and surrounding textures. Such dispersed responses make the detector more likely to produce unstable confidence scores near cluttered areas. RIEMamba instead concentrates the high-response regions around object bodies and their oriented boundaries, which is consistent with the cleaner boxes in the corresponding detection columns. In the scale-varying case in Fig. 8(b), small densely arranged objects and larger isolated objects coexist in the same crop. The compared backbones either miss part of the small objects or activate only local fragments of large objects, while RIEMamba preserves local details and wider contextual coverage at the same time. In the boundary-ambiguous case in Fig. 8(c), water-land transitions and weak object contours easily confuse background and foreground responses. RIEMamba produces sharper activations along object edges, reducing background diffusion around these weak boundaries. In the diverse-orientation case in Fig. 8(d), elongated objects arranged along different directions require consistent responses under rotation. The heatmaps show that RIEMamba better follows the object orientation rather than spreading along irrelevant linear structures.

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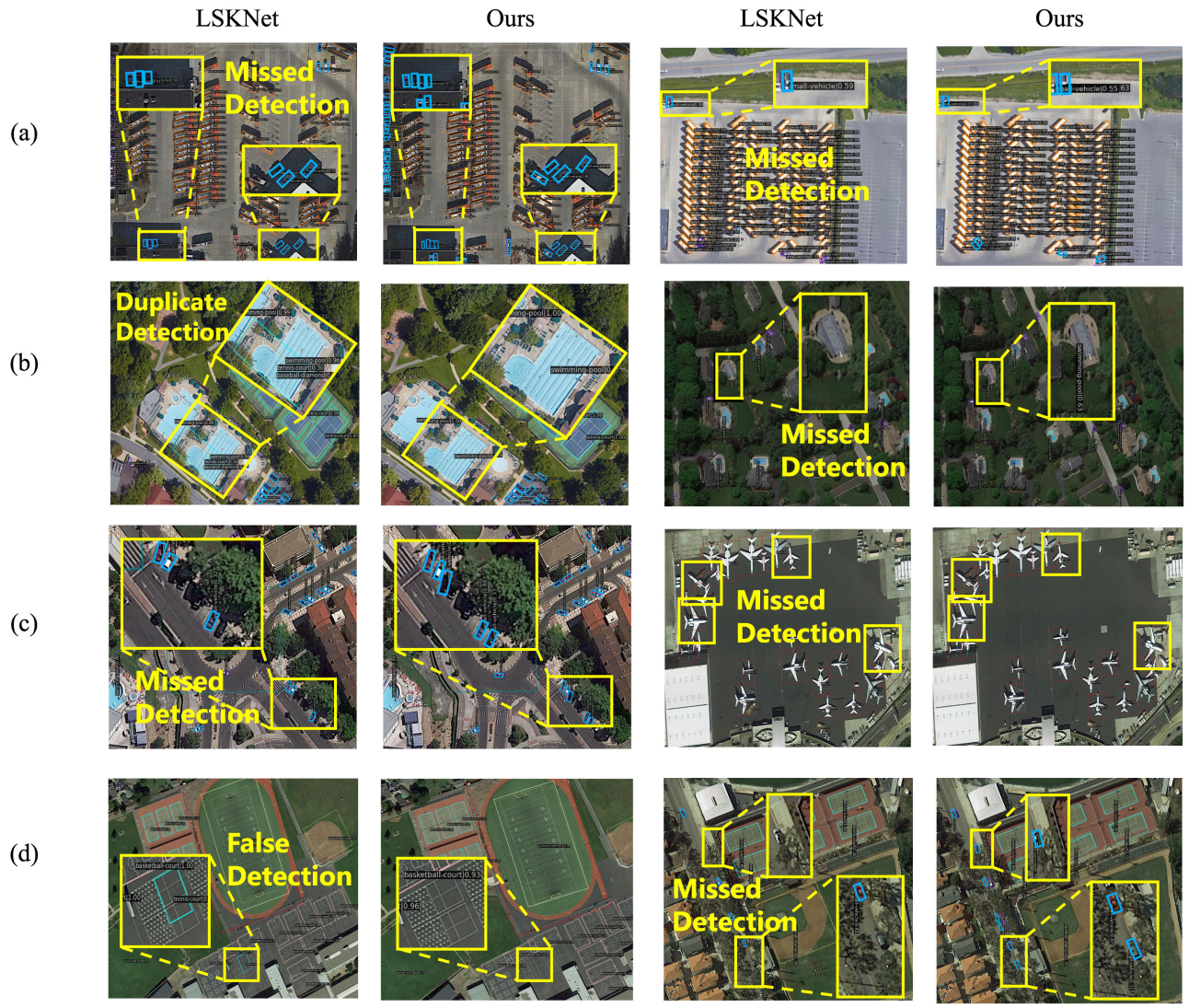


Fig. 10. Scene-level qualitative comparison between LSKNet-S and RIEMamba. In the figure, LSKNet denotes LSKNet-S and Ours denotes RIEMamba. Rows show (a) dense parking, (b) recreational areas, (c) transport hubs, and (d) sports grounds; dashed yellow boxes mark zoomed regions.

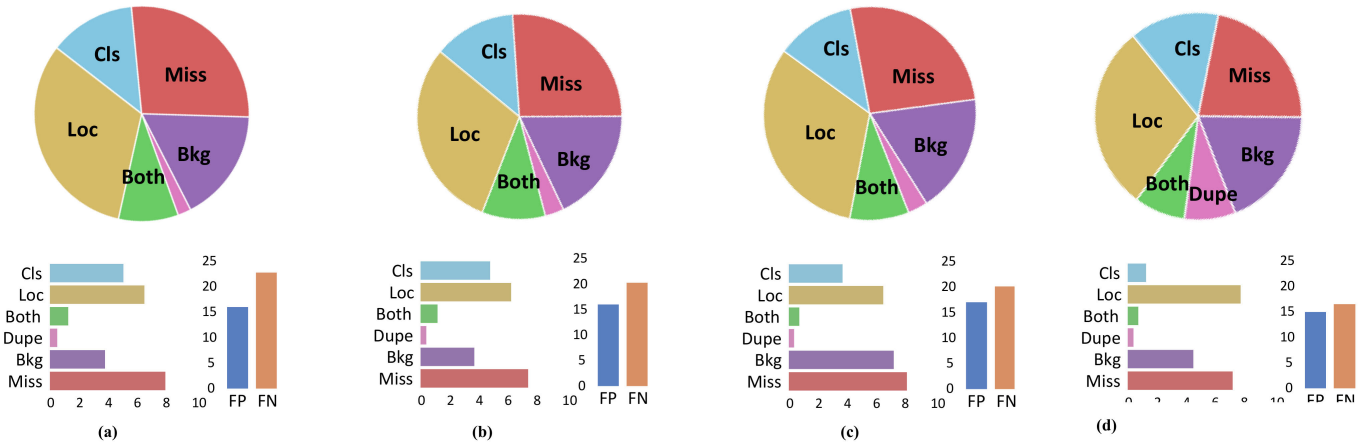


Fig. 11. Error analysis on the remote sensing test set using the TIDE [82] toolbox. (a) LSKNet-S. (b) PKINet-S. (c) MA3E. (d) RIEMamba (Ours).

The angular curves in Fig. 9 provide a complementary view of the same behavior. Across both DOTA-v1.0 and DIOR-R, RIEMamba keeps a more stable accuracy profile as the input angle changes, whereas LSKNet-T and LocalMamba-T show

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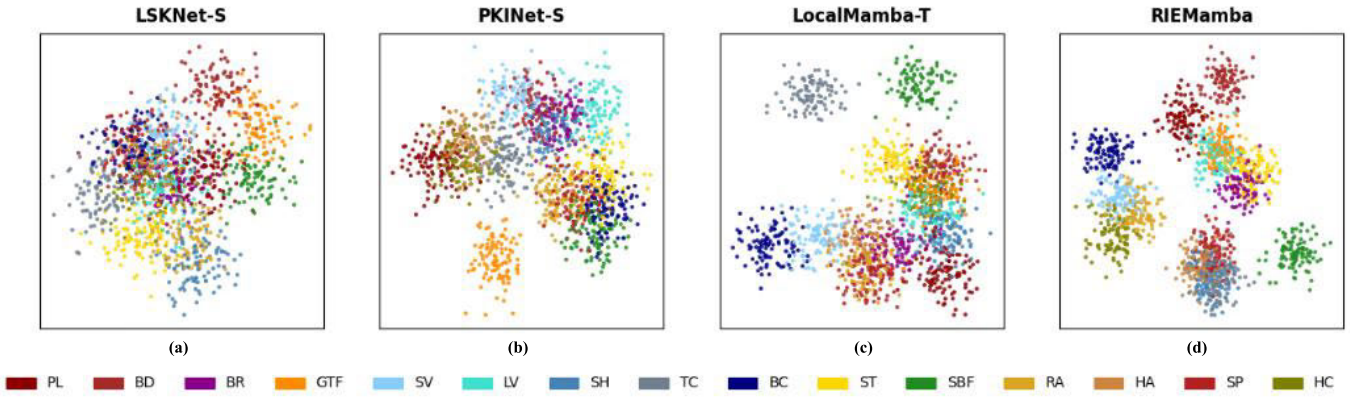


Fig. 12. t-SNE visualization of feature distributions on the DOTA-v1.0 dataset. (a) LSKNet-S. (b) PKINet-S. (c) LocalMamba-T. (d) RIEMamba (Ours). Different colors denote object categories.

larger drops at several oblique angles. This does not mean that the visualization alone proves the mechanism; rather, it explains where the quantitative gain comes from. RIEPDC improves edge-aware and rotation-consistent local responses, which helps suppress false activations around cluttered boundaries, while VMSS balances local detail and scene-level context, which helps recover small or partially covered objects. Therefore, the visual comparison, angular robustness, component ablation, and TIDE error decomposition support the same conclusion from different perspectives: RIEMamba improves oriented RSOD by making the backbone representation less sensitive to rotation, boundary ambiguity, and scale variation.

More specifically, Fig. 10 shows that the improvement is not limited to isolated examples. In the dense parking scenes in Fig. 10(a), the baseline misses several tightly arranged vehicles whose orientations and spacing are very similar to the surrounding parked cars, whereas RIEMamba recovers more complete rows of oriented boxes. In the recreational-area scenes in Fig. 10(b), LSKNet-S produces duplicate boxes on large court-like structures and also misses small buildings near cluttered vegetation; RIEMamba gives more compact detections because its boundary-aware responses suppress repeated activation on the same object. In the transport-hub scenes in Fig. 10(c), small targets near roads, docks, and aircraft parking regions are easily confused with background line structures, but RIEMamba detects more of these thin and oblique objects. In the sports-ground scenes in Fig. 10(d), the baseline introduces false positives around rectangular field markings and still misses nearby small targets, while RIEMamba better separates real objects from scene patterns. These cases complement the heatmap observations in Fig. 8: rotation-consistent edge modeling mainly reduces unstable responses on elongated or oblique objects, and scale-aware scanning helps retain small-object evidence without over-expanding activation on large background structures. This visual evidence also clarifies the detector-level trend reported later by TIDE. RIEMamba does not simply increase the number of boxes; it suppresses background responses while recovering targets that are weak, small, densely packed, or strongly rotated. Therefore, the qualitative examples are consistent with the reductions in missed detections, localization errors, and background false positives shown in Fig. 11.

3) Feature Separability: Fig. 12 visualizes instance-level feature distributions on DOTA-v1.0 with t-SNE. LSKNet-S, PKINet-S, and LocalMamba-T exhibit evident overlaps and scattered intra-class samples, whereas RIEMamba forms more compact and better separated clusters. This indicates that the proposed multi-scale orientation-consistent design strengthens feature discriminability under varying object orientations and scales.

V. Conclusion

In this article, we propose RIEMamba, a Mamba-based backbone for oriented RSOD that jointly addresses rotation sensitivity, edge ambiguity, and scale variation. By combining SO(2)-equivariant edge-aware pixel-difference modeling with adaptive dual-scale scanning, RIEMamba achieves state-of-the-art results on DOTA-v1.0 (80.11%/82.49% mAP), HRSC2016 (98.62% mAP under VOC 2012), and DIOR-R (68.55% mAP), with marginal overhead and consistent gains across six detection frameworks.

From a methodological perspective, RIEMamba focuses on two priors that are most critical for oriented RSOD: in-plane rotation-consistent boundary modeling and fixed dual-scale context aggregation. This design also defines its current boundary. RIEPDC does not explicitly model out-of-plane perspective changes caused by oblique views or terrain relief, and VMSS uses two preset scale branches rather than a fully learnable scale search. Future work may extend the geometric prior beyond SO(2) and make the scanning scale selection dynamically adaptive to more complex remote sensing scenes.

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